

OSI Layer Calculations and Network Communication Analysis

Assignment Submission

Course: Computer Networks

Topic: OSI Layer Numerical Problems and Analysis

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Introduction

The Open Systems Interconnection (OSI) model consists of seven layers that enable communication between networked devices. Each layer performs specific functions such as data segmentation, routing, framing, compression, encryption, synchronization, and application support. This assignment analyzes various numerical problems related to the OSI model and explains how different layers contribute to reliable and efficient network communication.

Question 1: Transport and Data Link Layer Analysis

Given Data

- File Size = 150 MB
- Bandwidth = 50 Mbps
- Segment Size = 1500 Bytes
- Data Link Overhead = 40 Bytes per frame

Formula

Number of Segments = File Size / Segment Size

Calculations

$$150 \text{ MB} = 150 \times 1024 \times 1024$$

$$= 157,286,400 \text{ Bytes}$$

Number of Segments

$$= 157,286,400 / 1500$$

$$\approx 104,858 \text{ Segments}$$

Number of Frames

$$= 104,858 \text{ Frames}$$

Total Overhead

$$= 104,858 \times 40$$

$$= 4,194,320 \text{ Bytes}$$

Total Data Transmitted

$$= 157,286,400 + 4,194,320$$

$$= 161,480,720 \text{ Bytes}$$

Convert to bits

$$= 161,480,720 \times 8$$

$$= 1,291,845,760 \text{ bits}$$

Transmission Time

$$= 1,291,845,760 / 50,000,000$$

$$\approx 25.84 \text{ seconds}$$

Explanation

The Transport Layer divides the file into manageable segments and provides error recovery through acknowledgments and retransmissions. The Data Link Layer encapsulates each segment into frames, adds addressing information, and performs error detection using CRC to ensure reliable communication.

Question 2: Sliding Window Protocol

Given Data

- Window Size = 6 Frames
- Total Frames = 48

Calculations

Number of Windows

$$= 48 / 6$$

$$= 8 \text{ Windows}$$

If one frame is lost in each window:

Retransmissions

$$= 8 \times 1$$

$$= 8 \text{ Frames}$$

Explanation

Flow control prevents the sender from overwhelming the receiver by limiting the number of frames sent before acknowledgment. This improves communication efficiency and avoids congestion.

Question 3: IP Fragmentation

Given Data

- Packet Size = 12,000 Bytes
- MTU = 1500 Bytes
- IP Header = 20 Bytes

Calculations

Maximum Data per Fragment

$$= 1500 - 20$$

$$= 1480 \text{ Bytes}$$

Number of Fragments

$$= \text{Ceiling}(12000 / 1480)$$

$$= 9 \text{ Fragments}$$

Header Overhead

$$= 9 \times 20$$

$$= 180 \text{ Bytes}$$

Explanation

The Network Layer fragments packets when they exceed the MTU. Each fragment carries identification information so the destination can correctly reassemble the original packet

Question 4: Sliding Window Protocol

Given Data

- Window Size = 6 Frames
- Total Frames = 48

Calculations

Transmission Windows

$$= 48 / 6$$

$$= 8 \text{ Windows}$$

Retransmissions

$$= 8 \text{ Frames}$$

Explanation

Flow control manages data transmission speed and ensures efficient utilization of network resources while preventing buffer overflow at the receiver.

Question 5: Session Layer Synchronization

Given Data

- Video File Size = 600 MB
- Checkpoint Interval = 60 MB
- Failure After = 390 MB

Calculations

Checkpoints Created

$$390 / 60$$

$$= 6 \text{ Checkpoints}$$

Last Checkpoint

$$= 360 \text{ MB}$$

Data to Retransmit

$$= 390 - 360$$

$$= 30 \text{ MB}$$

Explanation

Synchronization checkpoints enable communication recovery from the last saved state instead of restarting the entire transfer, saving bandwidth and time.

Question 6: Presentation Layer Compression and Encryption

Given Data

- Original File = 900 MB
- Compression = 40%
- Encryption Overhead = 5%

Calculations

Compressed Size

$$= 900 \times 0.60$$

$$= 540 \text{ MB}$$

Encrypted Size

$$= 540 \times 1.05$$

$$= 567 \text{ MB}$$

Reduction Percentage

$$= ((900 - 567) / 900) \times 100$$

$$= 37\%$$

Explanation

The Presentation Layer compresses data to reduce transmission time and encrypts data to maintain confidentiality and security during communication.

Question 7: FTP Data Transfer

Given Data

- Data Size = 3 GB
- Throughput = 120 Mbps
- Request Size = 3 MB

Calculations

3 GB

= 3072 MB

Number of Requests

= $3072 / 3$

= 1024 Requests

Convert Data to Bits

= 3072×8

= 24,576 Mb

Transmission Time

= $24,576 / 120$

= 204.8 seconds

Explanation

The Application Layer provides services such as file transfer, user authentication, error handling, and reliable communication through protocols like FTP.

Question 8: End-to-End Delay Analysis

Given Data

Application Delay = 4 ms

Transport Delay = 3 ms

Network Delay = 5 ms

Data Link Delay = 2 ms

Physical Delay = 1 ms

Propagation Delay = 18 ms

Transmission Delay = 12 ms

Calculations

Total Delay

$$= 4 + 3 + 5 + 2 + 1 + 18 + 12$$

$$= 45 \text{ ms}$$

Explanation

Each OSI layer performs specific processing functions. Together with transmission and propagation delays, they contribute to the total end-to-end delay experienced by a packet.

Question 9: Protocol Overhead and Efficiency

Given Data

- Useful Data = 80 MB
- Frame Size = 1600 Bytes
- Overhead = 80 Bytes

Calculations

Useful Data per Frame

$$= 1600 - 80$$

$$= 1520 \text{ Bytes}$$

80 MB

= 83,886,080 Bytes

Number of Frames

= 83,886,080 / 1520

≈ 55,188 Frames

Total Overhead

= 55,188 × 80

= 4,415,040 Bytes

Transmission Efficiency

= (1520 / 1600) × 100

= 95%

Explanation

Protocol overhead consumes additional bandwidth. Higher overhead reduces efficiency because more bytes are used for control information rather than useful data.

Question 10: Healthcare IoT Network Communication

Given Data

- Medical Record = 240 MB
- Compression = 30%
- Segment Size = 1400 Bytes
- Frame Overhead = 60 Bytes

Calculations

Compressed File Size

= 240 × 0.70

= 168 MB

Compressed Size in Bytes

$$= 168 \times 1024 \times 1024$$

$$= 176,160,768 \text{ Bytes}$$

Number of Segments

$$= 176,160,768 / 1400$$

$$\approx 125,829 \text{ Segments}$$

Total Overhead

$$= 125,829 \times 60$$

$$= 7,549,740 \text{ Bytes}$$

Final Data Transmitted

$$= 176,160,768 + 7,549,740$$

$$= 183,710,508 \text{ Bytes}$$

Explanation

The Presentation Layer compresses data, the Transport Layer segments data, the Data Link Layer frames data and adds error-checking information, and the Physical Layer transmits the data. Together, these layers ensure secure, reliable, and efficient communication.

Conclusion

This assignment demonstrated the practical application of OSI model concepts through numerical calculations involving segmentation, framing, fragmentation, synchronization, compression, encryption, flow control, and network efficiency. The results show how different layers cooperate to provide reliable, secure, and efficient communication across computer networks.

References

1. Forouzan, B. A. – Data Communications and Networking.
2. Andrew S. Tanenbaum – Computer Networks.

3. IEEE Networking Standards Documentation.
4. Computer Networks Course Notes.

Formatting Tips for Full Marks

- Use **Times New Roman, 12 pt** font.
- Use **1.5 line spacing**.
- Main headings: **Bold, 16 pt**.
- Subheadings: **Bold, 14 pt**.
- Add **page numbers**.
- Insert an **automatic Table of Contents**.
- Keep calculations and explanations separate under each question.
- Add a **cover page** and **references page**.

This structure aligns well with all rubric categories: Understanding, Explanation, Application, Organization, Accuracy, and Clarity.